Siamese metric learning for generalised few-shot recognition of rare viruses in transmission electron microscopy images

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Abstract

Background and Objective: Automatic classification of viruses in transmission electron microscopy (TEM) images requires many labelled examples of each virus, which rare and newly emerging viruses lack. The public TEM virus dataset contains 22 viruses, but its benchmark covers only the 14 most frequent. The eight rare viruses, with 5–25 training micrographs each, have not been studied. We examined whether a Siamese metric-learning encoder trained on the frequent viruses can recognise the rare ones from a few examples, without being trained on them.

Methods: A DenseNet201 encoder was trained with a supervised contrastive loss on the 14 base classes, and the rare classes were added at test time as class prototypes (generalised few-shot learning). The encoder was compared with a re-implemented benchmark classifier under identical, leakage-audited evaluation on the official split. All hyper-parameters were selected on validation, and differences were assessed with paired micrograph-level cluster bootstrap tests.

Results: On the 14 base classes, the encoder reached 94.79 % accuracy (macro-sensitivity 0.946, macro-specificity 0.9958) versus 93.91 % for the

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baseline, with no significant difference in accuracy for any seed pair. Confused rare classes were nearly linearly separable in the learned features. Replacing nearest-prototype classification with logistic regression on the frozen features raised the 22-class macro-F1 from 0.786 to 0.849, a significant gain for every checkpoint. With one example per rare virus, Siamese features gave higher rare-class accuracy than baseline features (37.3 % versus 30.8 %), but this advantage disappeared with three or more examples.

Conclusions: Siamese metric learning matches a strong classifier on this benchmark and is advantageous in the one-shot setting. For rare viruses, most of the attainable improvement comes from the decision rule rather than the representation.

Keywords: transmission electron microscopy, virus classification, few-shot learning, metric learning, rare classes

1. Introduction

Negative-stain transmission electron microscopy (TEM) can identify a virus without prior knowledge of what to look for, but interpreting the images requires scarce expertise. This has motivated automatic classification of virus particles in TEM images. Matuszewski and Sintorn released a public TEM virus dataset [1, 2]: 1,245 micrographs of 22 viruses, patches at 1 nm/pixel and an official, representative, micrograph-level train/validation/test split. Their benchmark uses the 14 most frequent classes, on which a fine-tuned DenseNet201 reached 93.1 % test accuracy. Eight classes were excluded because they had too few images or particles to train a classifier reliably.

Rare viruses are, however, the realistic diagnostic case: a newly emerg-

ing virus will always have few reference images. This is the setting of few-shot learning, and more precisely of generalised few-shot learning (GFSL), in which new classes must be recognised alongside the training classes [3]. Siamese metric learning is well suited to this setting. A shared-weight encoder maps patches of the same virus close together, and a new virus is represented by the mean embedding (prototype) of its few examples, without retraining. Because the benefit of few-shot methods can be overestimated when they are compared with weak baselines (Section 1.1), a credible evaluation must compare them with a strong classification baseline under identical conditions.

To our knowledge no published work has applied metric-learning few-shot methods to this dataset or evaluated its rare classes as few-shot classes. Our contributions are: (i) a GFSL protocol for all 22 classes on the official split, in which the rare classes are never used for training and every choice is made on validation; (ii) a Siamese supervised-contrastive encoder compared with a faithful re-implementation of the benchmark classifier under identical evaluation; (iii) a diagnosis showing that rare-class errors stem from the nearest-prototype decision rule rather than the representation, and an evaluation of classifiers on frozen features that addresses them; and (iv) full reporting of negative results, with code, logs and per-patch predictions released.

1.1. Related work

Virus classification on the TEM virus dataset. The benchmark of Matuszewski and Sintorn [1] compared a custom CNN with eight ImageNet-pretrained architectures on the 14-class split and also showed that an unrepre-

sentative split lowers performance substantially, which makes the official split the reference protocol. Later 14-class studies proposed ensembles of CNNs and vision transformers [4], p4-equivariant group convolutions that improve data efficiency [5], a compact dual-branch CNN fed with local-standard-deviation and discrete-cosine-transform images [6], and a DenseNet-based dual-track network with attention [7]. These studies reported accuracies between 84.49 % and 99.73 %, but under different protocols. They used a different train–test ratio [4], a reduced training set with results on the validation split [5], or a random split of the augmented patch set [6], and one did not report its protocol in the accessible text [7]. Their accuracies are therefore not directly comparable with each other or with the official benchmark. Two studies used all 22 classes: a dilated multilevel fused CNN [8] and a CNN–transformer hybrid evaluated on a stratified split [9], with accuracies of 89.89 % and 88.31 %, respectively. Both trained a closed-set classifier on every class. As far as we could determine, neither reported rare-class results on the official split or treated the rare viruses as few-shot classes.

Siamese networks, metric learning and few-shot baselines. Siamese networks learn a shared-weight encoder and compare inputs by distance [10, 11]. Prototypical networks extend this idea to N-way classification with class-mean prototypes [12]. Batch-based losses, namely the batch-hard triplet loss [13, 14] and the supervised contrastive (SupCon) loss [15], use every positive and negative pair in a class-balanced batch and train such encoders more efficiently than isolated pairs. Several studies have shown that a well-trained embedding with a simple classifier, such as nearest class mean after centring and normalisation [16] or logistic regression on

frozen features [17], rivals more elaborate meta-learning [18], and that the advantage of meta-learning shrinks under strong domain shift [19]. Richer decision rules on frozen features include class-specific Mahalanobis distances with shrunk covariances [20] and distribution calibration, which borrows feature statistics from related base classes [21].

Generalised few-shot learning and medical imaging. In GFSL, new classes must be recognised among the training classes [3]. The classification scores are typically biased towards the training classes, a bias that calibrated stacking counteracts [22]. In medical imaging, a systematic review of 80 few-shot studies found meta-learning to be the most common approach [23]. Few-shot recognition of viruses in TEM images, in which the rare classes are held out from training and evaluated on an official split, has, to our knowledge, not been studied.

2. Methods

2.1. Dataset and leakage control

We used the official patch folders of the dataset [2]: 256×256-pixel patches centred on annotated particles, after rescaling to 1 nm/pixel. The split assigns whole micrographs to train, validation or test (about 60/20/20 %) [1]. The 14 benchmark classes are the *base* classes. The eight excluded classes, which the dataset provides in the same split, are the *rare* classes (Table 1 and Fig. 1). The pre-augmented, class-balanced training set (736 patches per base class) was used only to re-implement the benchmark classifier, as in the original work.

Each patch was standardised to zero mean and unit variance [1] and its

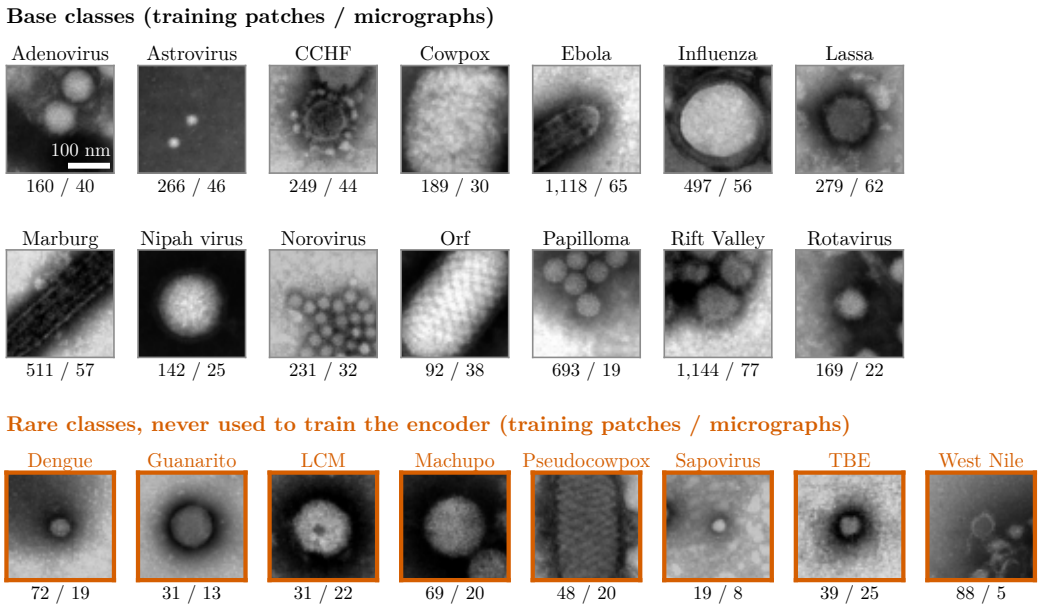


Figure 1: One example patch (256×256 pixels at 1 nm/pixel) of each virus from the official training split, with the number of training patches and micrographs. The eight rare classes (bottom, framed) are never used to train the encoder. Several are small, near-spherical particles of similar appearance, such as Dengue, Sapovirus and TBE. Scale bar, 100 nm (the same for all patches). Patches from the TEM virus dataset [2] (CC BY 4.0).

grey channel replicated to three channels. Patches were not resized. Before training we audited the data at pixel level: none of the 16,441 training and augmented-training patches is identical to any rotation or reflection of a validation or test patch, and no test patch matches a validation patch.

2.2. Siamese metric-learning encoder

The encoder (Fig. 2a) is an ImageNet-pretrained DenseNet201 [24, 25]. Global average pooling yields a 1,920-dimensional *backbone feature*, and a two-layer projection head (1,920→512→128 units) with L2 normalisation produces the embedding used by the loss. All patches pass through the same weights, so similarities are computed in a single shared space. The

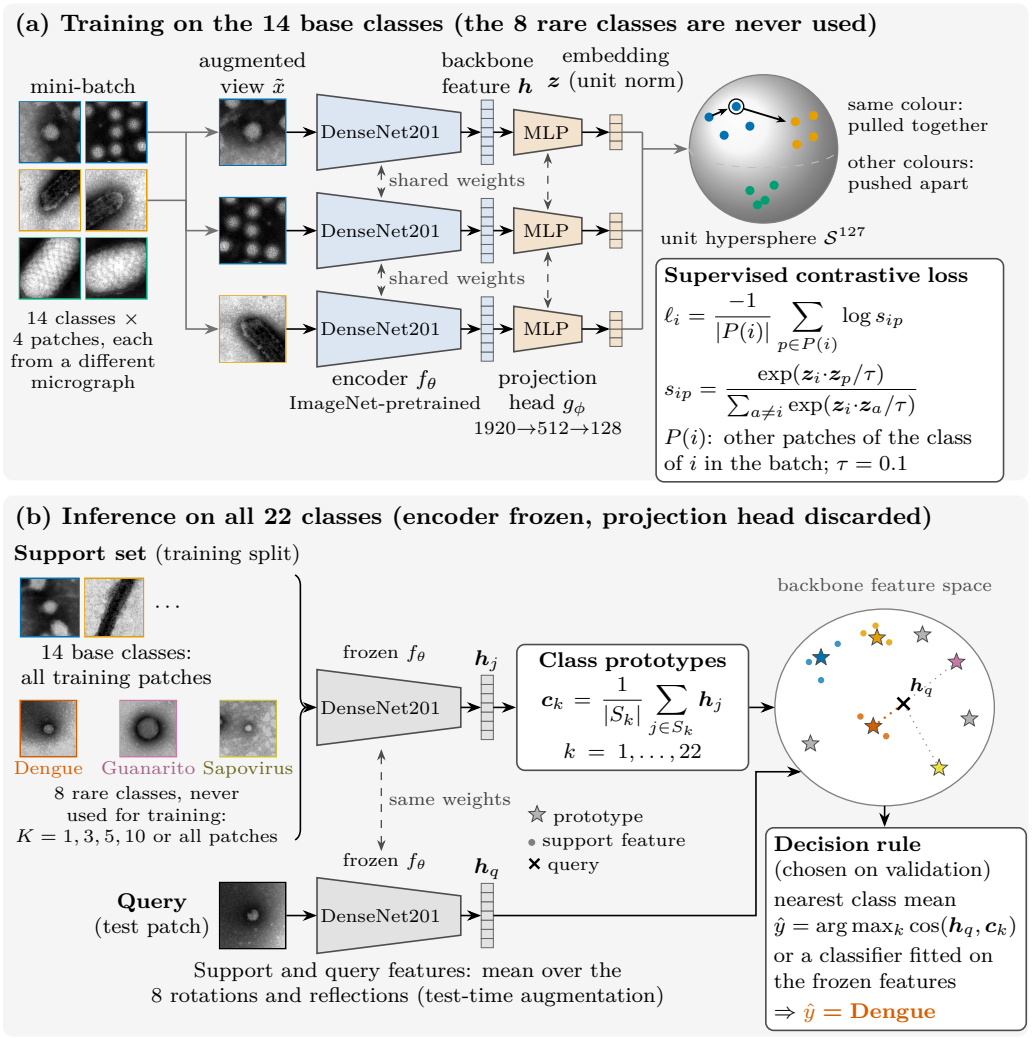


Figure 2: Overview of the method. (a) Training: a mini-batch contains four patches of each of the 14 base classes (three classes are shown; frame and point colours denote the virus). Each patch is randomly rotated or reflected and shifted by up to 8 pixels, and all patches pass through one DenseNet201 encoder with shared weights (drawn as parallel Siamese branches). A projection head maps the backbone feature $\boldsymbol{h}$ to a unit-norm embedding $\boldsymbol{z}$, and the supervised contrastive loss pulls together embeddings of the same virus and pushes apart those of different viruses. (b) Inference: the projection head is discarded, and the frozen encoder embeds the support patches of all 22 classes, including the eight rare classes never used for training, into class prototypes. A test patch is assigned by nearest class mean or by a classifier fitted on the frozen features, chosen on the validation split. Example patches are from the TEM virus dataset [2] (CC BY 4.0).

Table 1: Patches (micrographs) per class in the official split.

Class	Role	Train	Validation	Test
Adenovirus	base	160 (40)	44 (13)	86 (14)
Astrovirus	base	266 (46)	86 (15)	66 (15)
CCHF	base	249 (44)	100 (15)	86 (15)
Cowpox	base	189 (30)	72 (10)	59 (10)
Ebola	base	1,118 (65)	469 (22)	363 (22)
Influenza	base	497 (56)	281 (19)	170 (19)
Lassa	base	279 (62)	114 (21)	128 (21)
Marburg	base	511 (57)	174 (19)	173 (19)
Nipah virus	base	142 (25)	40 (8)	35 (8)
Norovirus	base	231 (32)	104 (11)	84 (11)
Orf	base	92 (38)	76 (13)	31 (13)
Papilloma	base	693 (19)	227 (6)	187 (6)
Rift Valley	base	1,144 (77)	414 (26)	392 (26)
Rotavirus	base	169 (22)	48 (7)	40 (7)
Dengue	rare	72 (19)	15 (6)	44 (7)
Guanarito	rare	31 (13)	10 (4)	7 (4)
LCM	rare	31 (22)	10 (7)	14 (8)
Machupo	rare	69 (20)	16 (6)	23 (7)
Pseudocowpox	rare	48 (20)	21 (7)	16 (7)
Sapovirus	rare	19 (8)	11 (3)	8 (3)
TBE	rare	39 (25)	12 (8)	11 (8)
West Nile	rare	88 (5)	66 (2)	34 (2)
Total		6,137 (745)	2,410 (248)	2,057 (252)

network was trained with the supervised contrastive loss [15]. Each batch contained several patches of every base class, and the patches of a class were drawn from different micrographs where possible. Augmentation used the eight exact rotations and reflections of the square and small translations, which preserve the physical scale. The network was optimised with AdamW [26]; all training settings are listed in Table 2. The checkpoint with the highest validation macro-F1 of nearest-prototype classification over the base classes was retained, and training stopped early when this score no longer improved (Fig. B.5). The rare classes were never seen during training or

model selection.

Pilot experiments in which four base classes were withheld from training showed that the backbone feature transfers to unseen classes whereas the 128-dimensional head does not (Appendix A). We therefore fixed the backbone feature as the representation for all experiments before evaluating the main models.

2.3. Baselines

The **baseline** is a re-implementation of the best benchmark model [1], based on its published configuration and code. It consists of DenseNet201 followed by two fully connected layers with dropout and a softmax output over the 14 base classes. It was trained with cross-entropy on the pre-augmented training set, and the checkpoint with the highest validation accuracy was retained (Table 2). The baseline and the Siamese encoder were each trained with three seeds. The pooled DenseNet201 features of the baseline also serve as an alternative backbone in every few-shot experiment, which isolates the effect of the training objective. A **fully supervised reference**, a DenseNet201 with a linear head trained with class-balanced sampling on all 22 classes, sees the rare classes during training and is reported for comparison only. Two loss ablations were trained once: the batch-hard triplet loss [14] and SupCon with an auxiliary cosine-classifier term [18].

2.4. Evaluation protocols

All results are reported on the official test split. For every model, including the baseline, each patch is represented by the mean of its features or class scores over its eight rotations and reflections (test-time augmentation),

Table 2: Training settings of the Siamese encoder and the baseline, read from the run configurations; they are identical for the three seeds of each model.

Setting	Siamese encoder	Baseline (benchmark classifier)
Backbone	DenseNet201, ImageNet-pretrained	DenseNet201, ImageNet-pretrained
Head	projection MLP 1,920→512→128, L2-normalised	2×FC 1,024, dropout 0.3, softmax
Loss	supervised contrastive, $\tau = 0.1$	cross-entropy
Training data	official training split	pre-augmented training split
Batch	14 classes × 4 patches (different micrographs)	12 patches
Optimiser	AdamW	Adam
Learning rate (backbone / head)	5×10^{-5} / 5×10^{-4}	1×10^{-5} / 1×10^{-5}
Weight decay	1×10^{-4}	0
Schedule	2 warm-up epochs, cosine decay	constant
Epochs	at most 40 (100 steps each), early stop after 12	50 (fixed)
Gradient clipping	5	none
Online augmentation	random rotation or reflection (8 exact orientations), shift of up to ±8 px	none
Checkpoint selection	validation macro-F1 (nearest class mean)	validation accuracy

computed with full-precision (fp32), deterministic inference. Three protocols were used. In the *closed-set* protocol (14 classes), test patches are assigned to the nearest class mean by cosine similarity, with prototypes computed from all training patches. The baseline uses its own softmax output. In the *GFSL* protocol (22 classes, Fig. 2b), prototypes of all classes are computed from the training split, and the 2,057 test patches are classified among all 22 classes. In the *K-shot GFSL* protocol, each rare class receives only $K \in \{1, 3, 5, 10\}$ training patches from different micrographs, drawn at random 100 times,

while the base classes keep all their training patches. We report accuracy, macro-F1, base-class accuracy and the rare-class mean per-class accuracy. The last measure is preferred to pooled accuracy because the rare-class test sets range from 7 to 44 patches. Results for episodes and unknown-virus detection are given in Appendix B.

2.5. Classifiers on frozen features

To test whether the rare-class errors stem from the representation or from the decision rule, four classifiers were fitted on the frozen backbone features of every checkpoint, without modifying the encoder. These were (i) the nearest class mean, (ii) L2-regularised logistic regression with class-balanced weights [17], (iii) a Mahalanobis classifier whose class covariances are shrunk towards the pooled covariance in a 64–256-dimensional principal-component space, following Simple CNAPS [20], and (iv) distribution calibration, which samples synthetic features for each rare class from the statistics of its nearest base classes [21]. Each classifier was fitted on the training split, and its hyper-parameters were chosen by the 22-class macro-F1 on the validation split. The chosen setting was then applied once to the test split. In the K-shot setting, every classifier and checkpoint was evaluated on the same 30 support draws. Before any test result was examined, we fixed the rule that the classifier with the highest validation macro-F1 is the one reported as the improvement. All four classifiers are nevertheless reported. In a separability diagnostic, a binary logistic regression was trained on the training patches of each of eight confusable pairs and evaluated on their validation patches.

2.6. Statistics

Confidence intervals (95 %) come from a stratified micrograph-level cluster bootstrap that resamples micrographs within classes [27]. Paired comparisons on identical test patches use a paired cluster bootstrap of the difference in accuracy and macro-F1, together with an exact McNemar test [28]. Because the McNemar test treats patches as independent, it is anti-conservative when patches share micrographs. The conclusions therefore rest on the cluster bootstrap. The Siamese encoder and the baseline are compared for all 3×3 seed pairs, and K-shot comparisons are paired over identical support draws. Values over seeds are given as mean $\pm$ standard deviation.

2.7. Implementation

Experiments used PyTorch 2.11 and scikit-learn on one NVIDIA RTX 3060 (12 GB). The analysis code was written with the assistance of an AI tool (Claude, Anthropic) and was checked with 23 automated tests, including comparisons of the loss functions and evaluation metrics against independent reference implementations. Training took 19–28 min per seed for the Siamese encoder and about 134 min per seed for the baseline.

3. Results

3.1. 14-class benchmark

The Siamese encoder performed on par with the baseline (Table 3). It was more accurate in all nine seed pairs, but none of the differences in accuracy was statistically significant, and only one pair differed significantly in macro-F1. Every seed of both models exceeded the published accuracy of 93.1 %.

Table 3: 14-class official test split (1,900 patches, 206 micrographs). Macro-averaged precision, sensitivity (recall) and specificity. Values over seeds are the mean, with the SD below it. [a]Without test-time augmentation.

Model	Acc. (%)	Macro-F1	Prec.	Sens.	Spec.	Micrograph acc. (%)
DenseNet201, published [1][a]	93.10	0.921	0.926	0.921	–	–
Baseline: benchmark model re-trained here (3 seeds)	93.91 ±0.54	0.930 ±0.007	0.927 ±0.006	0.939 ±0.013	0.9952 ±0.0005	95.95 ±0.74
Siamese SupCon, backbone feature (3 seeds)	94.79 ±0.21	0.943 ±0.004	0.944 ±0.005	0.946 ±0.003	0.9958 ±0.0001	96.60 ±0.49
Ablation: batch-hard triplet (1 seed)	93.58	0.929	0.933	0.928	0.9948	96.12
Ablation: SupCon + cosine CE (1 seed)	94.37	0.938	0.947	0.934	0.9954	96.60
Siamese 128-d head (not pre-specified)	95.72 ±0.35	0.952 ±0.003	0.955 ±0.005	0.951 ±0.003	0.9965 ±0.0003	96.93 ±0.28

Because test-time augmentation and implementation details differ from the original study, the re-trained baseline rather than the published figure is the appropriate comparator. The 128-dimensional projection head scored slightly higher on the base classes (Table 3). However, it was not the pre-specified representation, its advantage over the baseline was not significant in the seed-0 comparison, and it fails on unseen classes (Section 3.2).

3.2. 22-class generalised few-shot recognition

With nearest-prototype classification over all 22 classes, the Siamese and baseline features reached the same macro-F1 (Table 4). The Siamese features were more accurate overall, mainly because they classified the base classes better, whereas the baseline features recognised the rare classes slightly bet-

Table 4: 22-class GFSL on the official test split (2,057 patches, 252 micrographs) with nearest-class-mean classification; rare classes enter only as prototypes. Values over seeds are the mean, with the SD below it.

Features	Acc. (%)	Macro-F1	Base acc. (%)	Rare mean-class acc. (%)
Siamese backbone feature (3 seeds)	89.56 ± 0.36	0.786 ± 0.008	91.8 ± 0.3	74.5 ± 2.7
Baseline backbone feature (3 seeds)	87.57 ± 0.38	0.785 ± 0.008	89.4 ± 0.6	77.4 ± 1.7
Siamese 128-d head	89.35 ± 0.18	0.693 ± 0.013	94.1 ± 0.1	39.1 ± 5.1
Ablation: batch-hard triplet	88.67	0.779	90.7	76.4
Ablation: SupCon + cosine CE	89.35	0.794	91.2	79.4
Fully supervised reference, trained on all 22 classes	*93.10*	*0.897*	*95.1*	*81.8*

ter. The accuracy advantage of the Siamese features was significant in only two of the nine seed pairs. For seed 0, the patch-level McNemar test indicated a significant difference ($p = 0.00064$) although the cluster bootstrap interval included zero, which illustrates why the conclusions rest on the cluster bootstrap. The 128-dimensional head collapsed on the rare classes, with a rare-class mean accuracy of only 33–43 %. A large gap remained to the fully supervised reference, which had been trained on the rare classes. The errors were concentrated in a few rare viruses. In every seed, most Dengue test patches (32–35 of 44) were assigned to Sapovirus.

3.3. Rare classes are separable

Every confusable pair was close to linearly separable in the learned features, with a validation AUROC of at least 0.94 (Table 5). Nevertheless, the two-class nearest-mean rule performed far worse than a linear boundary

Table 5: Pairwise separability on the validation split (binary logistic regression trained on training patches). Values are means over the three seeds of each encoder. Val. patches: number of validation patches of the two classes.

	Siamese features			Baseline features	
Pair	AUROC	Log. reg. acc.	Nearest-mean acc.	AUROC	Val. patches
Dengue vs Sapovirus	1.000	1.00	0.91	1.000	15/11
Dengue vs TBE	0.974	0.95	0.80	0.981	15/12
Dengue vs Astrovirus	1.000	0.95	0.87	1.000	15/86
Sapovirus vs Norovirus	0.993	0.99	0.99	0.993	11/104
Sapovirus vs Astrovirus	0.957	0.87	0.49	0.920	11/86
Guanarito vs Lassa	0.986	0.96	0.85	0.989	10/114
Guanarito vs Machupo	0.958	0.86	0.67	0.948	10/16
LCM vs Machupo	0.979	0.91	0.87	0.990	10/16

for several pairs. For Sapovirus versus Astrovirus, it was close to chance, whereas logistic regression separated the pair well. The information needed to recognise the rare viruses is therefore present in the representation but is not exploited by an isotropic distance to the class mean. Because the rare-class validation sets are small (10–16 patches), these values should be regarded as indicative.

3.4. Classifiers on frozen features

The pre-declared selection rule chose logistic regression for six of the seven checkpoints, including all three Siamese seeds (Table 6). The only exception was baseline seed 1 (shrunk Mahalanobis classifier). For the Siamese encoder, logistic regression raised the 22-class macro-F1 from 0.786 to 0.849 and also improved the base-class accuracy. The gain over the nearest class mean was significant for all seven checkpoints. Among the weakest rare classes, the F1 score of Dengue rose from 0.12 to 0.34 (Appendix Table B.8).

Fig. 3 shows that the improvement came mainly from higher precision. Averaged over the eight rare classes, precision increased from 0.49 to 0.69, whereas recall changed less (from 0.74 to 0.80). With the nearest class mean, many patches of other classes were assigned to some rare classes. For example, the precision for Sapovirus was only 0.11. Dengue remained the most difficult class. Even with logistic regression, 52 % of its test patches were assigned to Sapovirus, compared with 77 % with the nearest class mean. The shrunk Mahalanobis classifier reached a similar macro-F1, but its validation scores were almost identical across settings, so its test result depends on near-ties. We therefore report it as a secondary result.

With all rare-class training data available, the Siamese training objective gave no advantage, because logistic regression on the baseline features reached a similar macro-F1 (Table 6). The difference between the two feature sets was significant in only one of the nine seed pairs, in favour of the baseline.

Classifiers that estimate more parameters required more data. With a single example per rare class, logistic regression reached a rare-class mean accuracy of only 18.0 %, compared with 36.9 % for the nearest class mean (Table 6). Distribution calibration, in contrast, improved on the nearest class mean in this setting, significantly for all six Siamese and baseline checkpoints. With distribution calibration, the Siamese and baseline features performed similarly.

3.5. Few-shot regime

With one training patch per rare virus, the Siamese features gave a higher rare-class mean accuracy than the baseline features, 37.3 % versus 30.8 %

Table 6: 22-class GFSL on the official test split with classifiers on frozen backbone features; hyper-parameters selected on validation. Values over seeds are the mean, with the SD below it. $K = 1$, $K = 5$: rare mean-class accuracy with only K training patches per rare class, averaged over 30 support draws (the same draws for every classifier and checkpoint), so these values differ slightly from Fig. 4, which averages 100 draws. Superscripts: number of seeds for which the classifier had the highest validation macro-F1.

Classifier	Val. F1	Acc. (%)	Macro-F1	Rare (%)	K=1 (%)	K=5 (%)
Siamese features						
Nearest class mean	0.758 ± 0.019	89.56 ± 0.36	0.786 ± 0.008	74.5 ± 2.7	36.9 ± 2.6	65.6 ± 1.5
Logistic regression$^{3/3}$	0.851 ± 0.011	92.72 ± 0.27	0.849 ± 0.009	79.6 ± 1.0	18.0 ± 0.8	58.9 ± 0.7
Shrunk Mahalanobis	0.838 ± 0.008	92.56 ± 0.69	0.861 ± 0.008	75.0 ± 4.8	25.3 ± 4.8	47.5 ± 8.7
Distribution calibration	0.754 ± 0.012	89.95 ± 0.65	0.776 ± 0.008	73.8 ± 0.6	43.5 ± 2.4	64.4 ± 2.2
Baseline features						
Nearest class mean	0.766 ± 0.020	87.57 ± 0.38	0.785 ± 0.008	77.4 ± 1.7	29.7 ± 2.0	67.6 ± 1.8
Logistic regression$^{2/3}$	0.872 ± 0.012	92.55 ± 0.67	0.855 ± 0.013	79.5 ± 1.8	14.6 ± 1.4	57.0 ± 2.0
Shrunk Mahalanobis$^{1/3}$	0.859 ± 0.017	91.33 ± 0.29	0.846 ± 0.001	76.1 ± 1.8	15.5 ± 1.2	49.6 ± 2.1
Distribution calibration	0.754 ± 0.017	88.59 ± 0.71	0.766 ± 0.010	76.9 ± 6.1	45.8 ± 3.1	69.1 ± 5.8

(Fig. 4). When paired over identical support draws (30 per seed pair), the difference was significant in all nine seed pairs. From three examples onwards, the baseline features performed equally well or better (Appendix Table B.10).

4. Discussion

Under an identical, leakage-audited protocol, a Siamese supervised-contrastive encoder with a nearest-prototype classifier performed on par with a re-implementation of the strongest published benchmark model on

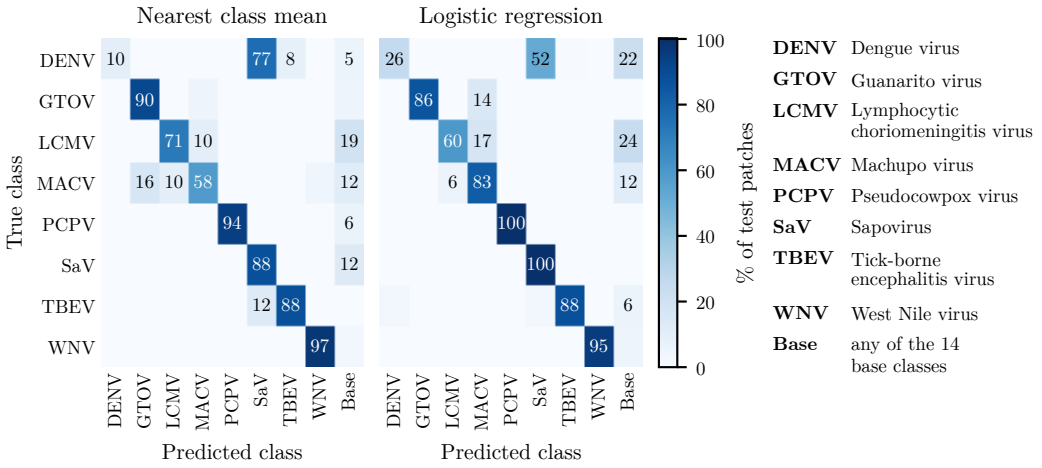


Figure 3: Where the rare-class test patches go, for Siamese features with nearest class mean (left) and with logistic regression (right). Rows: true rare class; columns: predicted rare class, or any of the 14 base classes pooled (Base). Virus abbreviations are given in the key on the right. Counts are summed over the three seeds and normalised per row (%); values below 5 % are not printed.

the 14 base classes. Its features also allow new virus classes to be added without retraining. In the most data-scarce setting, with a single reference patch per new virus classified by the nearest prototype, the Siamese encoder recognised rare viruses better than features learned with cross-entropy, consistently across all seed pairs. This is the setting that motivates few-shot learning for diagnostic TEM, because the first image of an emerging virus can be added as a prototype immediately.

The advantage did not persist with three or more examples, in rare-class-only episodes, when distribution calibration was used, or once a linear classifier was fitted on all available rare-class data. With nearest-prototype classification over all 22 classes, the two feature sets also reached the same macro-F1. These findings agree with previous reports that the quality of the embedding, rather than the metric-learning objective, dominates few-shot

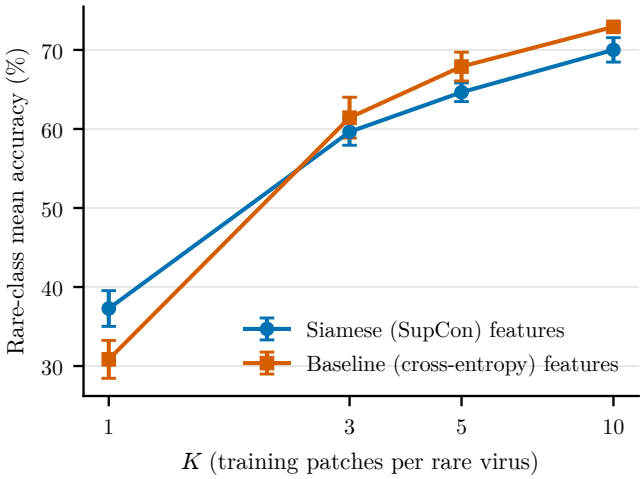


Figure 4: K-shot GFSL with nearest class mean: rare-class mean accuracy when each rare class has only K training patches (from different micrographs), while the 14 base classes keep all training patches. Mean ± SD over three seeds per encoder; each seed averages 100 random support draws.

performance [18, 17]. They also indicate that claims of superiority for metric-learning approaches on this dataset should be tested against an identically evaluated classification baseline.

The most practically relevant result concerns the rare classes, which previous 22-class studies did not analyse. Dengue, Sapovirus, Guanarito and LCM were poorly recognised by prototype classification, although they are nearly linearly separable from the classes with which they are confused. The distance to a class mean ignores differences in class spread, which is plausible for rare classes whose few micrographs differ in staining and particle density. A validation-selected logistic regression recovered much of the gap to the fully supervised reference (macro-F1 of 0.849 versus 0.897) without retraining the encoder. However, this classifier requires enough examples. With one to five examples per class, the nearest class mean or distribution

calibration should be preferred.

Published 22-class results were obtained by training on all classes with different or unreported splits [8, 9], so they are not directly comparable with ours. For reference, the fully supervised model trained on all classes with our pipeline reached 93.10 % accuracy on the official split.

Limitations. The rare classes have only 2–8 test micrographs each, so the per-class F1 scores of the smallest classes, such as Sapovirus and Guanarito with fewer than ten test patches each, are highly uncertain. The ablations were trained with a single seed. The hyper-parameters of the frozen-feature classifiers were chosen on small rare-class validation sets, which adds selection variance, as the Mahalanobis results show. Finally, we studied one backbone and one dataset, and the unknown-virus detection reported in Appendix B is not yet adequate for deployment.

5. Conclusions

To our knowledge, this is the first generalised few-shot evaluation of all 22 classes of the public TEM virus dataset on its official split, with the rare classes never used for training. A Siamese metric-learning encoder performs on par with a strong re-implemented classifier on the benchmark. With nearest-prototype classification, it recognises rare viruses better when only one reference patch is available, but not when more examples are available. Most of the attainable improvement on rare viruses comes from replacing nearest-prototype classification with a validation-selected linear classifier on frozen features, for either training objective. The protocol, code and per-patch predictions are released for future comparisons.

Declaration of competing interests

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and code availability

The TEM virus dataset is publicly available [2]. Code, configurations, training logs, evaluation outputs and per-patch predictions are available at `https://github.com/hrid0yyy/tem-virus-siamese-fewshot` [TODO before submission: repository is still private – make it public or change this URL]; trained checkpoints are available as release assets.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic) in order to assist with writing the analysis code and with drafting and editing

the manuscript text. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Appendix A. Pilot experiments

Four base classes were withheld from encoder training (F2: Astrovirus, Influenza, Papilloma, Rotavirus; F3: Marburg, Orf, Adenovirus, CCHF) and used as unseen classes, with micrograph-disjoint episodes (600 episodes, 15 queries per class). Without training, ImageNet features reached only 66.0 % nearest-mean accuracy on the 14 classes (88.1 % with a linear probe). After training, the backbone feature transferred to the unseen classes, whereas the 128-dimensional head did not; in the F3 GFSL test head accuracy on the unseen classes was 26.9 % (Table A.7).

Table A.7: Pilot experiments (accuracy in %, or harmonic mean of base and unseen accuracy).

Experiment	Frozen ImageNet	Trained, backbone feature	Trained, 128-d head
F2: 4-way 5-shot on held-out classes	86.9	96.9 (+10.0 ± 0.4)	84.8
F3: 4-way 5-shot on held-out classes	91.2	96.0 (+4.8 ± 0.3)	93.1
F3: GFSL harmonic mean (10 base + 4 held-out)	0.676	0.843	0.419

Appendix B. Additional results

The training curves of the three Siamese and three baseline runs are shown in Fig. B.5. The Siamese runs stopped early once the validation macro-F1

Table B.8: Test F1 of the weakest rare classes. Test patches: Dengue 44, Sapovirus 8, Guanarito 7, LCM 14, Machupo 23.

Features, classifier	Dengue	Sapovirus	Guanarito	LCM	Machupo
Siamese, nearest class mean	0.12	0.20	0.43	0.49	0.62
Siamese, logistic regression	0.34	0.35	0.77	0.64	0.80
Siamese, shrunk Mahalanobis	0.44	0.26	0.82	0.78	0.81
Siamese, distribution calibration	0.15	0.22	0.47	0.48	0.62
Baseline, nearest class mean	0.12	0.20	0.51	0.51	0.68
Baseline, logistic regression	0.30	0.35	0.86	0.68	0.76
Baseline, shrunk Mahalanobis	0.38	0.27	0.77	0.65	0.75
Baseline, distribution calibration	0.16	0.20	0.47	0.48	0.61

Table B.9: 8-way episodes over the rare classes (micrograph-disjoint support and query, 600 episodes) and unknown-virus detection (rare-class test patches as unknown; score = maximum cosine similarity to the 14 base prototypes).

Features	1-shot (%)	5-shot (%)	AUROC	FPR at 95 % TPR (%)
Siamese (3 seeds)	65.0 ± 1.3	78.2 ± 1.0	0.883 ± 0.003	58.2 ± 3.1
Baseline (3 seeds)	65.7 ± 0.8	80.3 ± 1.0	0.881 ± 0.015	60.7 ± 5.8
SupCon + cosine CE	65.6	79.0	0.899	52.9
Batch-hard triplet	62.9	77.1	0.868	70.7

had not improved for 12 epochs. The baseline was trained for the fixed 50 epochs of the original configuration. Its validation accuracy fluctuated from epoch to epoch, so the selected checkpoint (highest validation accuracy) varies between seeds.

Table B.10: Numerical values of Fig. 4: rare-class mean accuracy (%) with K training patches per rare class, nearest class mean, mean ± SD over three seeds (100 draws per seed).

K	Siamese features	Baseline features
1	37.3 ± 2.3	30.8 ± 2.4
3	59.6 ± 1.7	61.4 ± 2.6
5	64.6 ± 1.2	67.9 ± 1.8
10	70.0 ± 1.5	72.9 ± 0.3

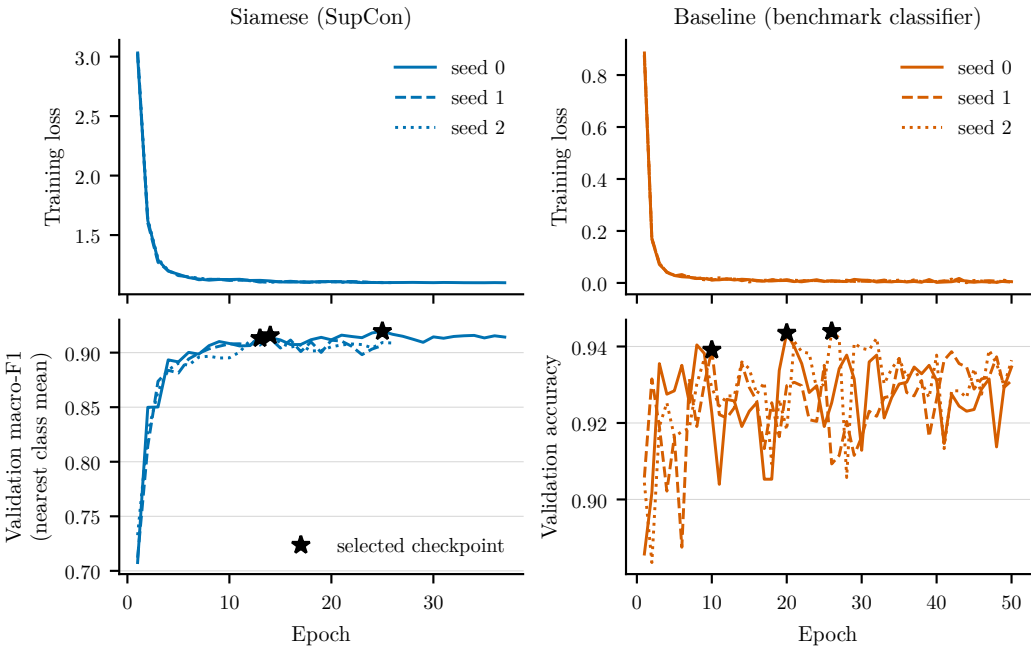


Figure B.5: Training loss (top) and the validation metric used for checkpoint selection (bottom) per epoch for the three seeds of each encoder. Stars mark the selected checkpoints.

References

[1] D. J. Matuszewski, I.-M. Sintorn, TEM virus images: Benchmark dataset and deep learning classification, Computer Methods and Programs in Biomedicine 209 (2021) 106318. doi:10.1016/j.cmpb.2021.106318.

[2] D. J. Matuszewski, I.-M. Sintorn, TEM virus dataset, version 3, Mendeley Data (2021). doi:10.17632/x4dwwfwtw3.3.

[3] S. Gidaris, N. Komodakis, Dynamic few-shot visual learning without forgetting, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2018.

[4] L. Nanni, A. Loreggia, L. Barcellona, S. Ghidoni, Building ensemble of deep networks: Convolutional networks and transformers, IEEE Access (2023). doi:10.1109/ACCESS.2023.3330442.

[5] K. Bengtsson Bernander, I.-M. Sintorn, R. Strand, I. Nyström, Classification of rotation-invariant biomedical images using equivariant neural networks, Scientific Reports 14 (2024) 14995. doi:10.1038/s41598-024-65597-x.

[6] N. Sikder, M. A.-M. Khan, A. K. Bairagi, M. Masud, J. J. Tiang, A.-A. Nahid, Heterogeneous virus classification using a functional deep learning model based on transmission electron microscopy images, Scientific Reports 14 (2024) 28954. doi:10.1038/s41598-024-80013-0.

[7] A. S. Bisht, R. Karthik, A. Ajay, A. K. Singh, Virus-FusionNet: an explainable deep learning-based approach for virus classification from

TEM images using DenseNet and lightweight hierarchical semi-local feature extraction network, Engineering Research Express 7 (3) (2025) 035242. doi:10.1088/2631-8695/adf411.

[8] M. Usman, H. Sultan, J. S. Hong, S. G. Kim, R. Akram, H. A. H. Gondal, M. H. Tariq, K. R. Park, Dilated multilevel fused network for virus classification using transmission electron microscopy images, Engineering Applications of Artificial Intelligence 138 (2024) 109348. doi:10.1016/j.engappai.2024.109348.

[9] M. M. Hoque, M. K. Islam, M. M. R. Moon, A. R. Akand, M. H. Al-amin, H. M. Azrof, A comparative and hybrid study of CNN and transformer models for multi-class virus classification in transmission electron microscopy, arXiv preprint arXiv:2606.14722, preprint (2026).

[10] J. Bromley, I. Guyon, Y. LeCun, E. Säckinger, R. Shah, Signature verification using a "Siamese" time delay neural network, in: Advances in Neural Information Processing Systems 6, 1993, pp. 737–744.

[11] G. Koch, R. Zemel, R. Salakhutdinov, Siamese neural networks for one-shot image recognition, in: ICML Deep Learning Workshop, 2015.

[12] J. Snell, K. Swersky, R. Zemel, Prototypical networks for few-shot learning, in: Advances in Neural Information Processing Systems 30, 2017, arXiv:1703.05175.

[13] F. Schroff, D. Kalenichenko, J. Philbin, FaceNet: A unified embedding for face recognition and clustering, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2015.

[14] A. Hermans, L. Beyer, B. Leibe, In defense of the triplet loss for person re-identification, arXiv preprint arXiv:1703.07737 (2017).

[15] P. Khosla, P. Teterwak, C. Wang, A. Sarna, Y. Tian, P. Isola, A. Maschinot, C. Liu, D. Krishnan, Supervised contrastive learning, in: Advances in Neural Information Processing Systems 33, 2020.

[16] Y. Wang, W.-L. Chao, K. Q. Weinberger, L. van der Maaten, SimpleShot: Revisiting nearest-neighbor classification for few-shot learning, arXiv preprint arXiv:1911.04623 (2019).

[17] Y. Tian, Y. Wang, D. Krishnan, J. B. Tenenbaum, P. Isola, Rethinking few-shot image classification: A good embedding is all you need?, in: European Conference on Computer Vision (ECCV), 2020.

[18] W.-Y. Chen, Y.-C. Liu, Z. Kira, Y.-C. F. Wang, J.-B. Huang, A closer look at few-shot classification, in: International Conference on Learning Representations (ICLR), 2019.

[19] Y. Guo, N. C. Codella, L. Karlinsky, J. V. Codella, J. R. Smith, K. Saenko, T. Rosing, R. Feris, A broader study of cross-domain few-shot learning, in: European Conference on Computer Vision (ECCV), 2020, arXiv:1912.07200.

[20] P. Bateni, R. Goyal, V. Masrani, F. Wood, L. Sigal, Improved few-shot visual classification, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020.

[21] S. Yang, L. Liu, M. Xu, Free lunch for few-shot learning: Distribution

calibration, in: International Conference on Learning Representations (ICLR), 2021.

[22] W.-L. Chao, S. Changpinyo, B. Gong, F. Sha, An empirical study and analysis of generalized zero-shot learning for object recognition in the wild, in: European Conference on Computer Vision (ECCV), 2016.

[23] E. Pachetti, S. Colantonio, A systematic review of few-shot learning in medical imaging, arXiv preprint arXiv:2309.11433 (2023). doi:10.48550/arXiv.2309.11433.

[24] G. Huang, Z. Liu, L. van der Maaten, K. Q. Weinberger, Densely connected convolutional networks, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2017.

[25] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, L. Fei-Fei, ImageNet: A large-scale hierarchical image database, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2009.

[26] I. Loshchilov, F. Hutter, Decoupled weight decay regularization, in: International Conference on Learning Representations (ICLR), 2019.

[27] B. Efron, R. J. Tibshirani, An Introduction to the Bootstrap, Chapman & Hall, 1993.

[28] Q. McNemar, Note on the sampling error of the difference between correlated proportions or percentages, Psychometrika 12 (2) (1947) 153–157.